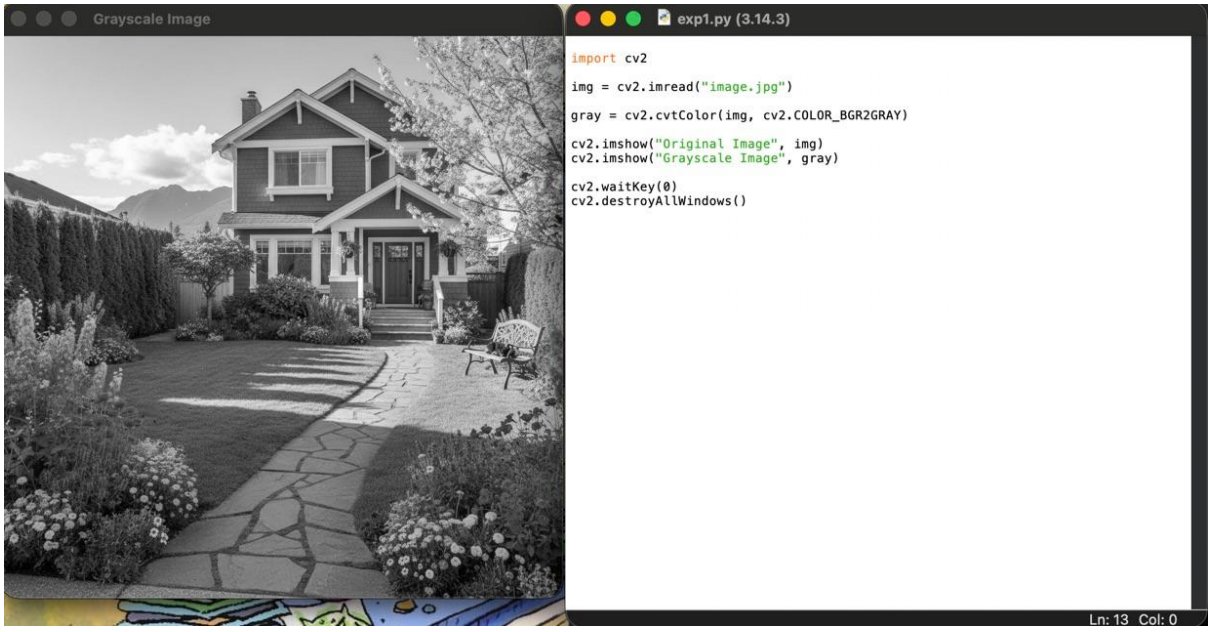
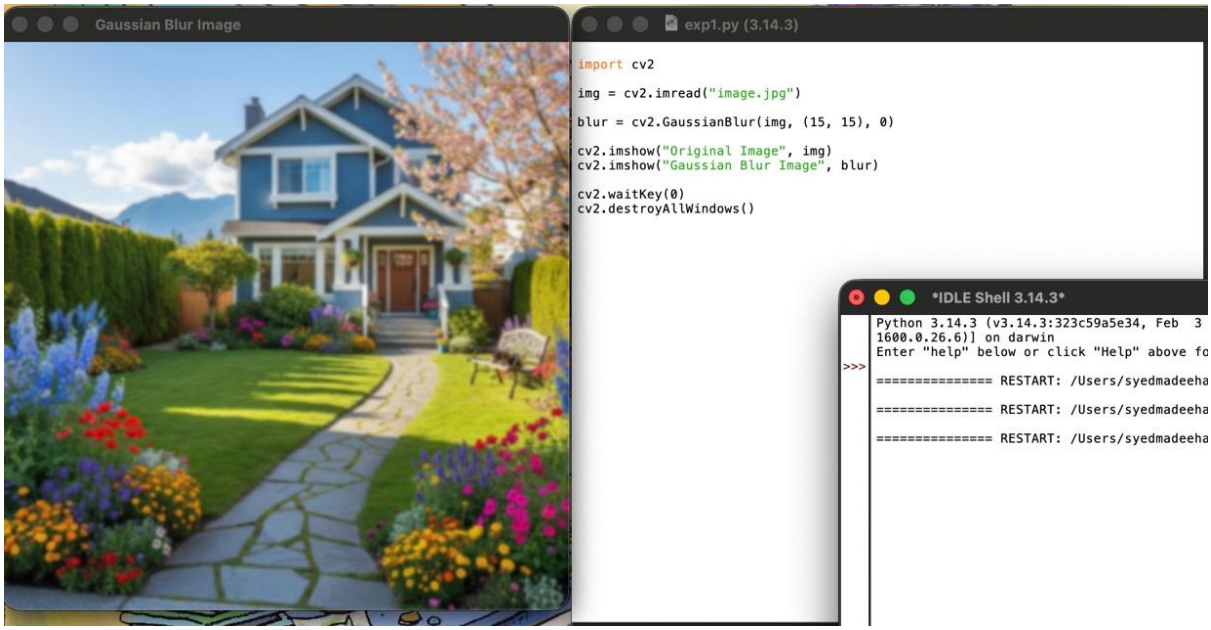
ITA0517-COMPUTER VISION

LAB EXPERIMENTS

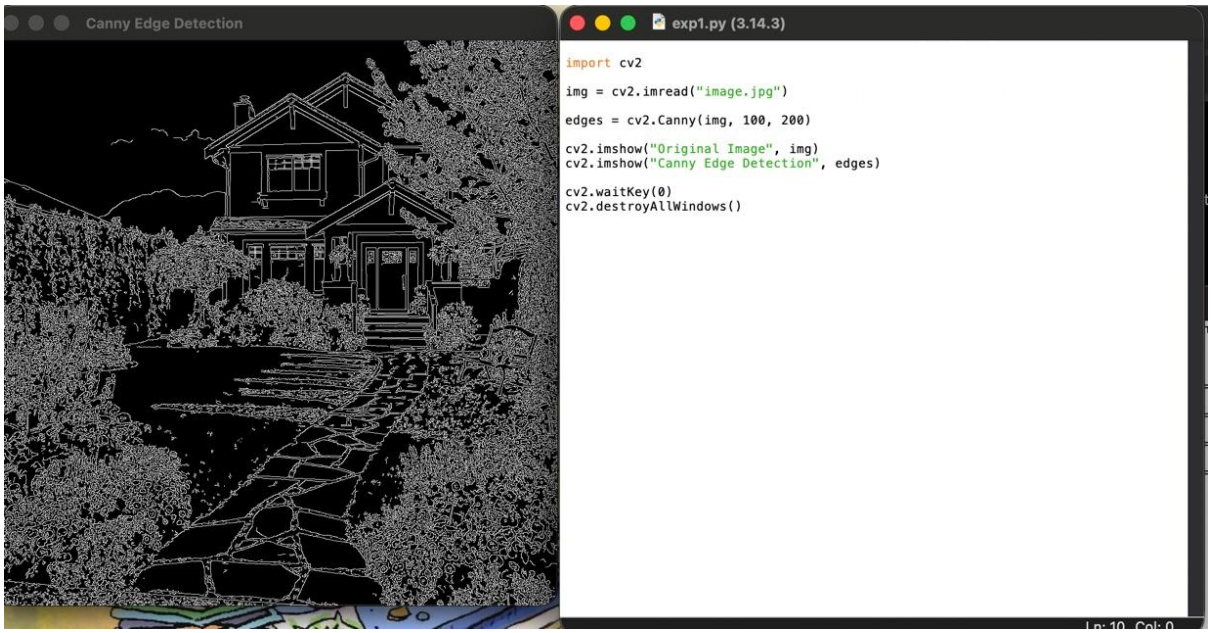
1.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Gray-scale.



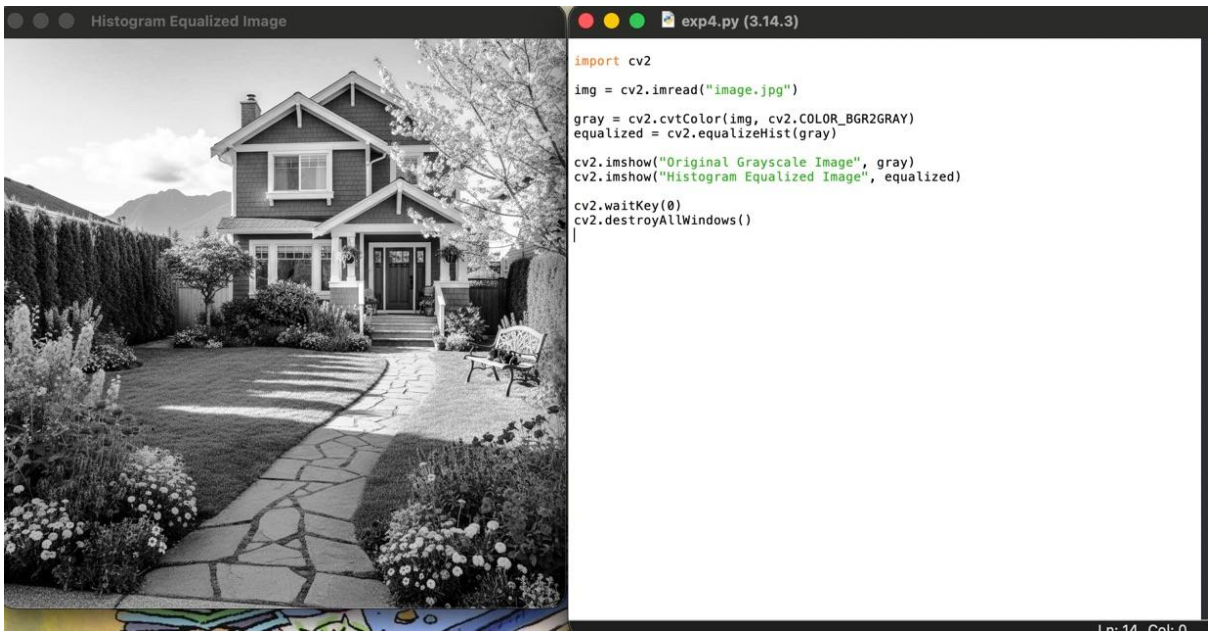
2. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Blur using Gaussian Blur



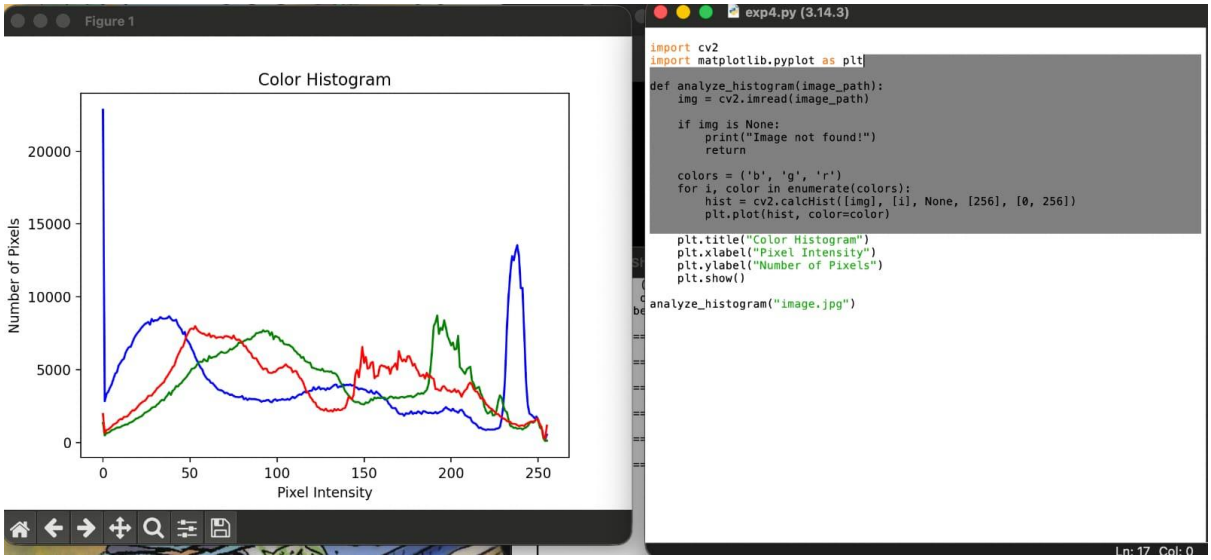
3.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.



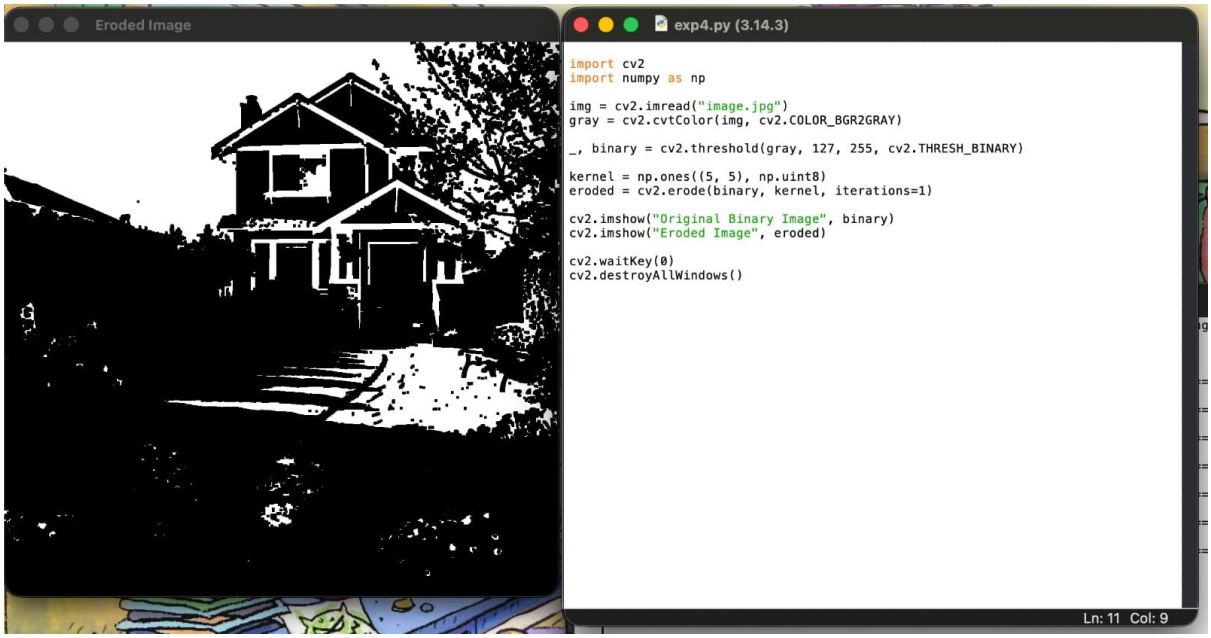
4.Implement histogram equalization on the given image and compare it with the original image using Open CV.



5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.



6.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.



7.Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

CV_Experiment7.py - C:/Users/chakr/OneDrive/Desktop/CV_Experiment7.py (3.12.0)

File Edit Format Run Options Window Help

```
import cv2

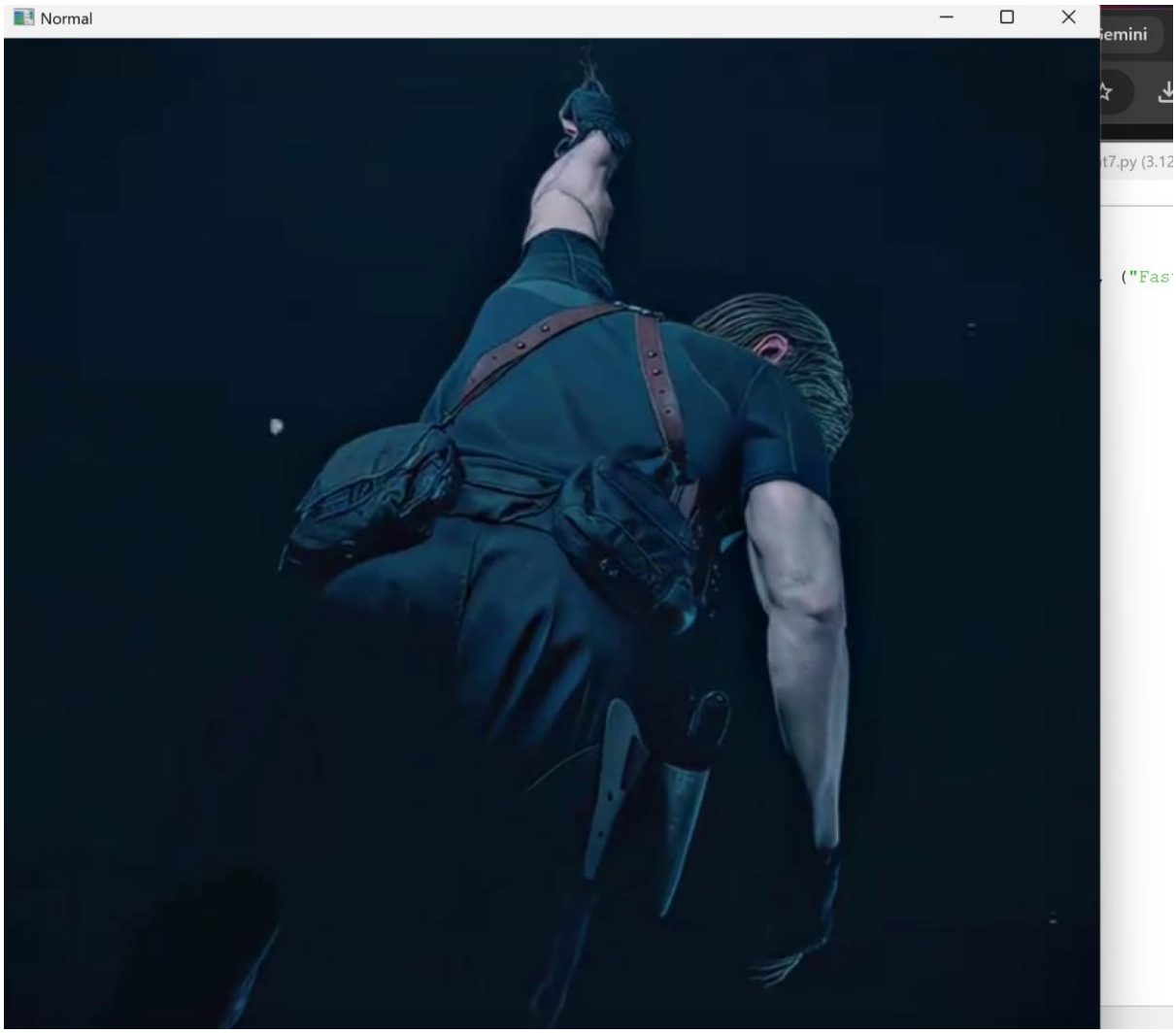
for name, delay in [("Normal", 30), ("Slow", 100), ("Fast", 1)]:
    cap = cv2.VideoCapture("video.mp4")

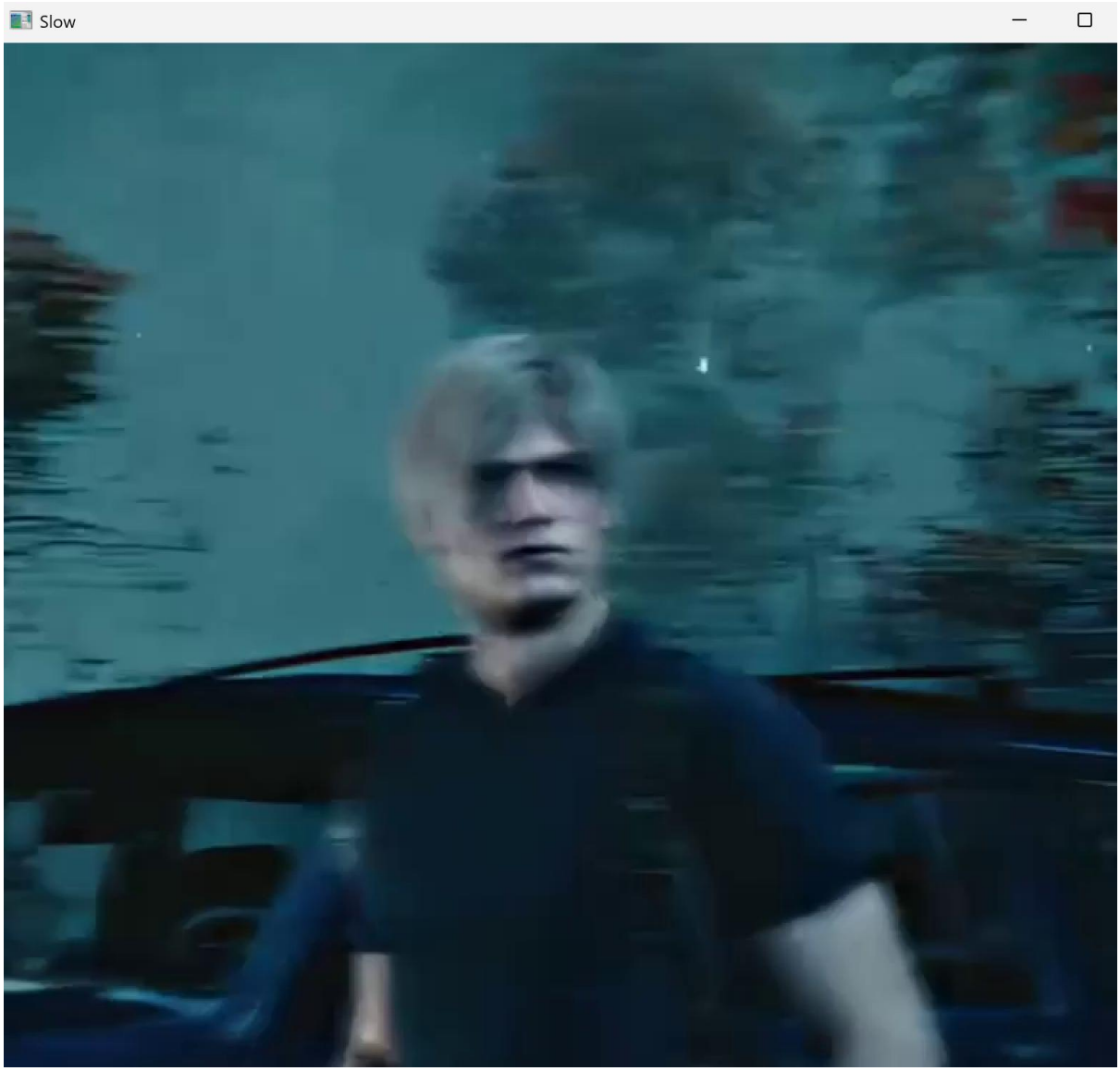
    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break

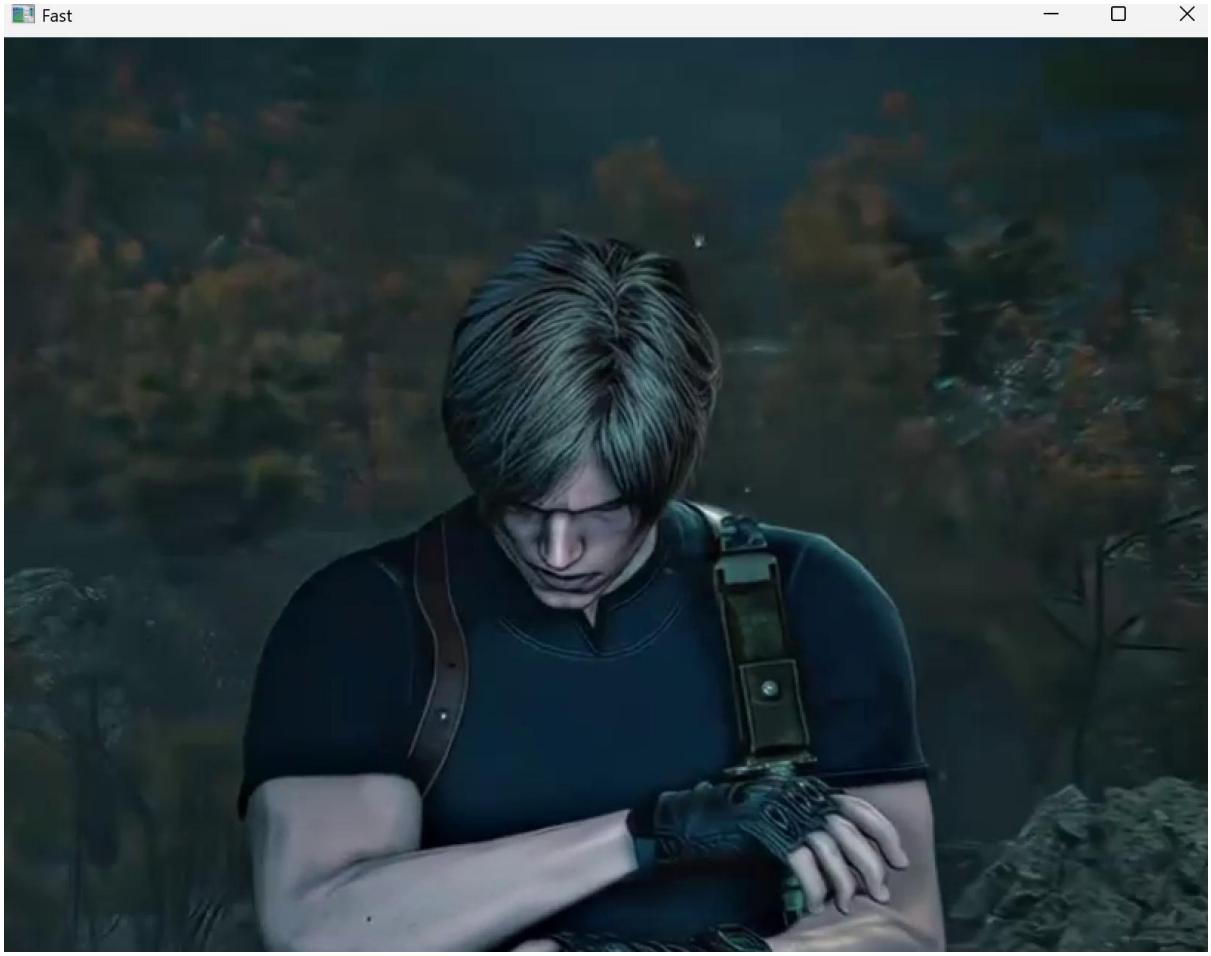
        cv2.imshow(name, frame)

        if cv2.waitKey(delay) & 0xFF == ord('q'):
            break

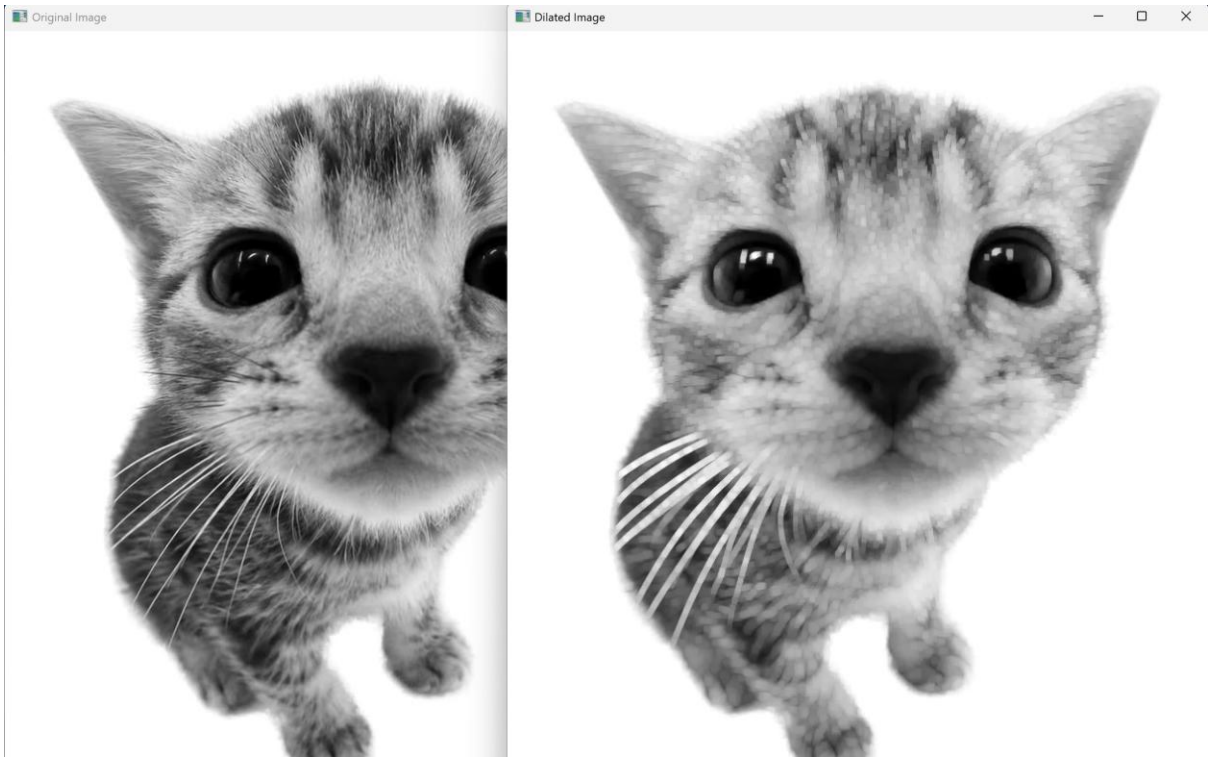
    cap.release()
    cv2.destroyAllWindows()
```




Slow



8.Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function



experiment8.py - C:/Users/chakr/Downloads/experiment8.py (3.12.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

kernel = np.ones((5, 5), np.uint8)
result = cv2.dilate(img, kernel, iterations=1)

cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

9.Implement the image scaling techniques to resize the images to both bigger and small sizes

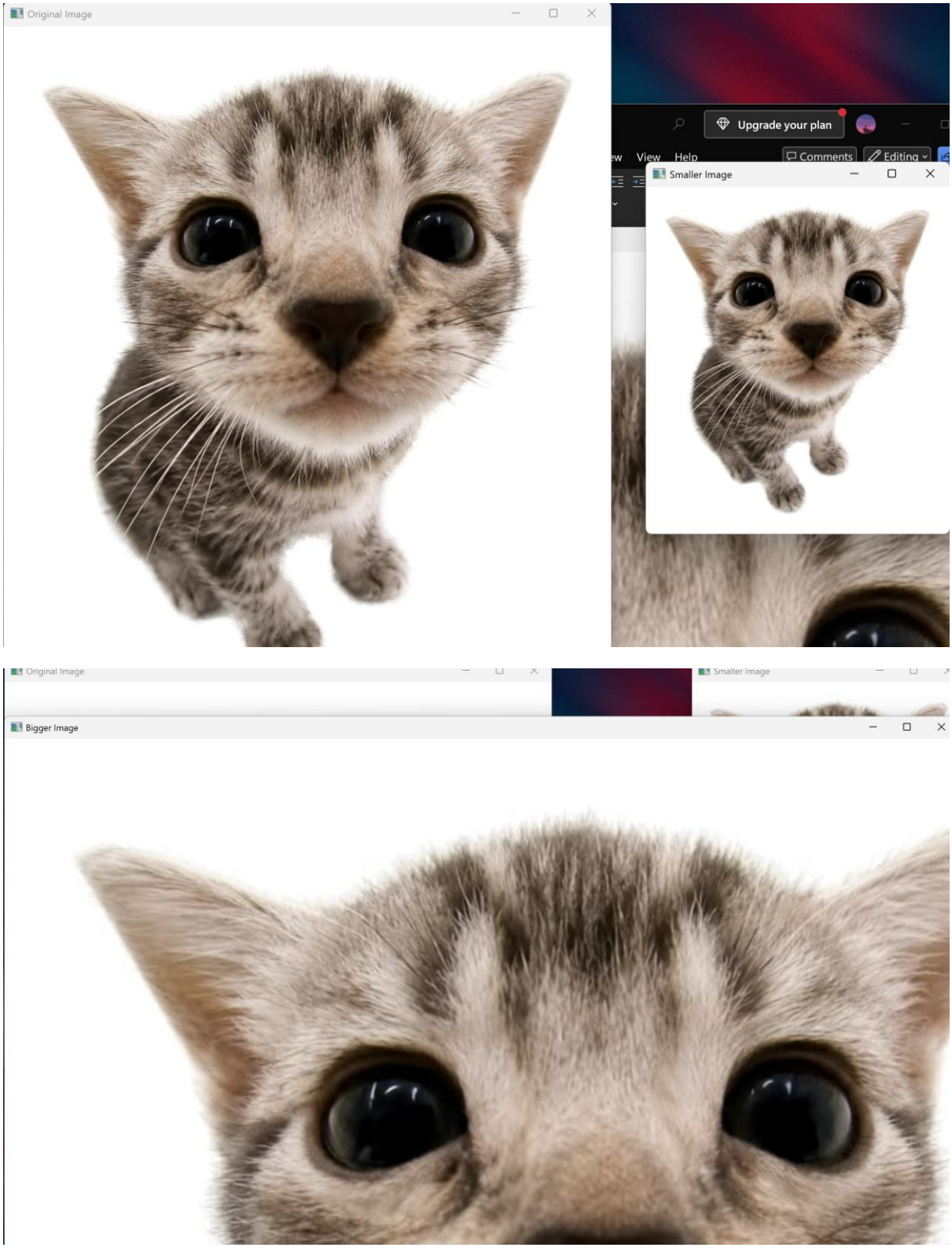
```
import cv2

img = cv2.imread("image.jpg.jpeg")

if img is not None:
    big = cv2.resize(img, None, fx=2, fy=2)
    small = cv2.resize(img, None, fx=0.5, fy=0.5)

    cv2.imshow("Original Image", img)
    cv2.imshow("Bigger Image", big)
    cv2.imshow("Smaller Image", small)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



10.Perform a 90-degree rotation clockwise along the y-axis for the given image.

```
File  Edit  Format  Run  Options  Window  Help

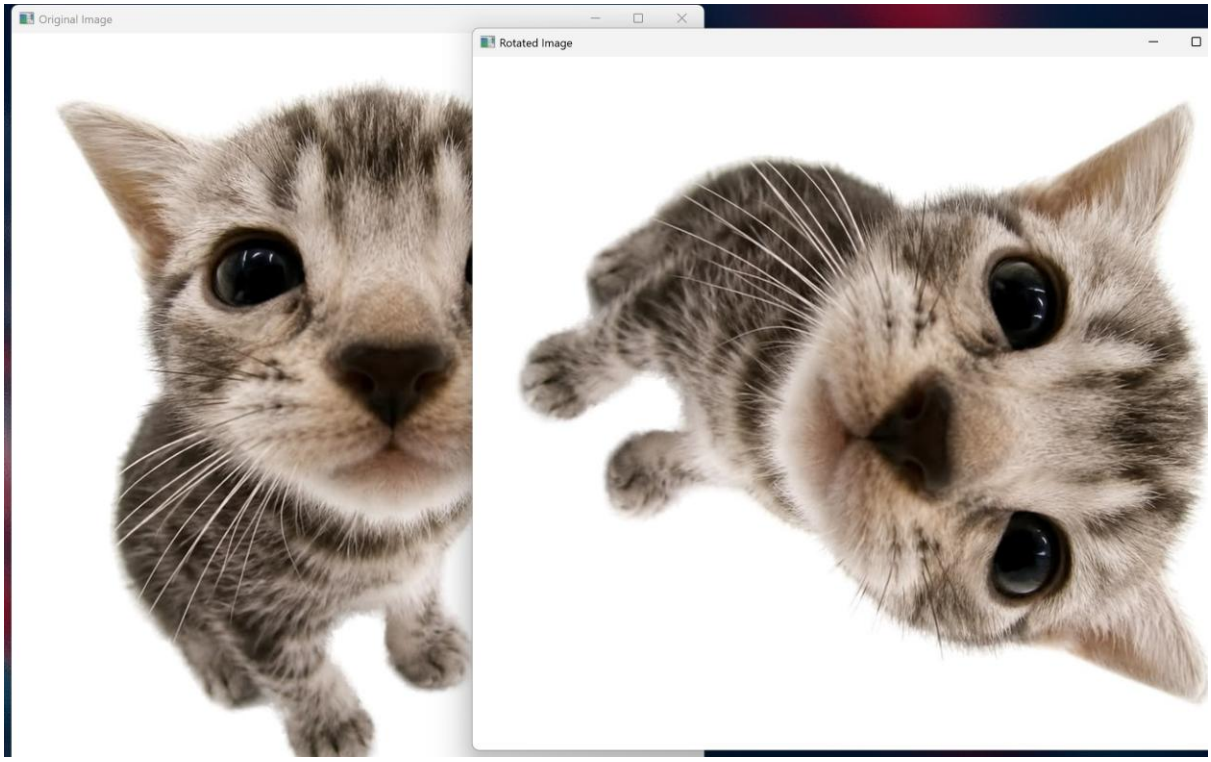
import cv2

img = cv2.imread("image.jpg.jpeg")

if img is not None:
    rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

    cv2.imshow("Original Image", img)
    cv2.imshow("Rotated Image", rotated)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



11 .Perform a 180-degree rotation clockwise along the y-axis for the given image

File Edit Format Run Options Window Help

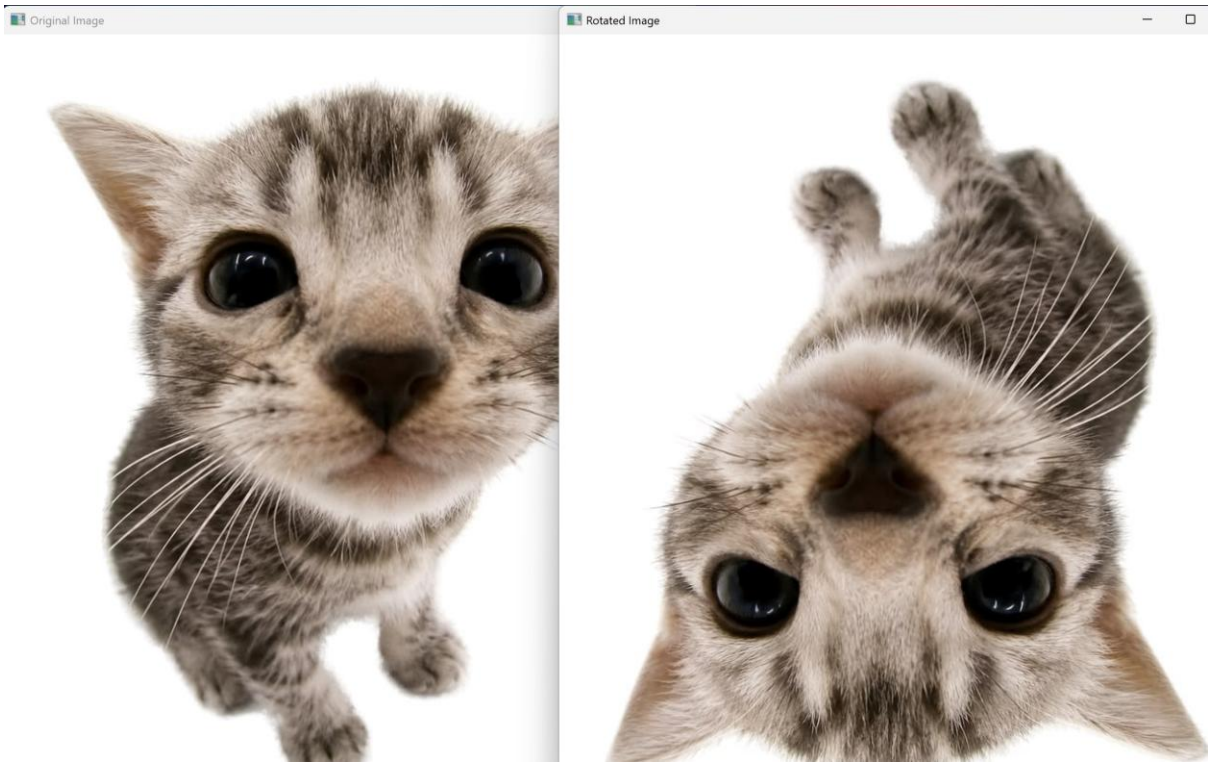
```
import cv2

img = cv2.imread("image.jpg.jpg")

if img is not None:
    rotated = cv2.rotate(img, cv2.ROTATE_180)

    cv2.imshow("Original Image", img)
    cv2.imshow("Rotated Image", rotated)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



12.Perform a 270-degree rotation clockwise along the y-axis for the given image.

File Edit Format Run Options Window Help

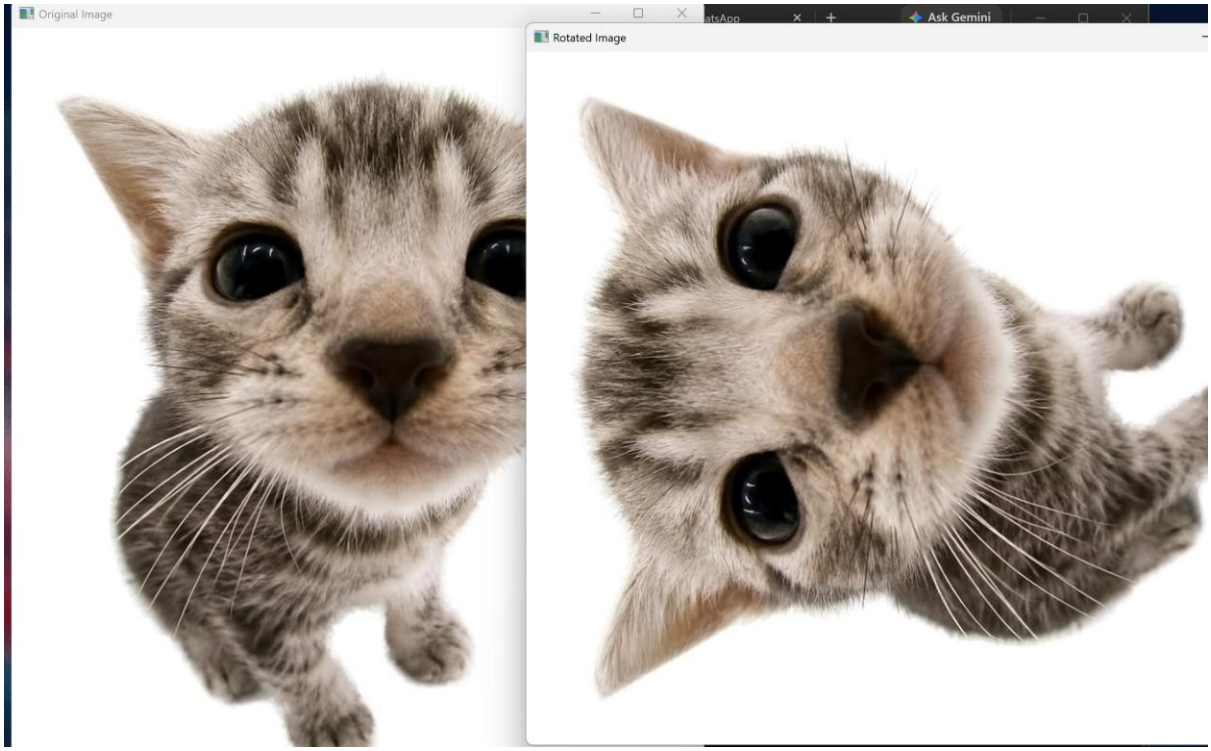
```
import cv2

img = cv2.imread("image.jpg.jpeg")

if img is not None:
    rotated = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

    cv2.imshow("Original Image", img)
    cv2.imshow("Rotated Image", rotated)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



13.Perform Affine Transformation on the given image using python and Open CV

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg")

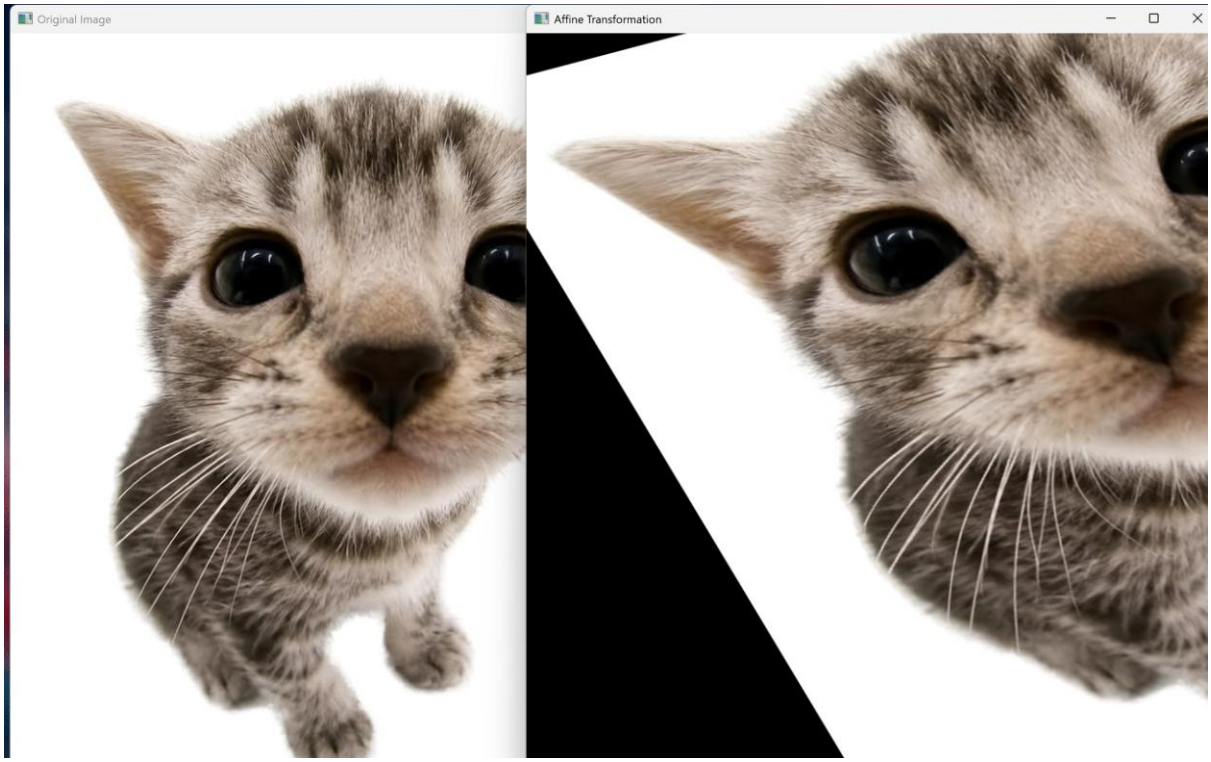
if img is not None:
    rows, cols = img.shape[:2]

    src = np.float32([[50, 50], [200, 50], [50, 200]])
    dst = np.float32([[10, 100], [200, 50], [100, 250]])

    M = cv2.getAffineTransform(src, dst)
    result = cv2.warpAffine(img, M, (cols, rows))

    cv2.imshow("Original Image", img)
    cv2.imshow("Affine Transformation", result)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



14.Perform Perspective Transformation on the given image using python and Open CV.

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg")

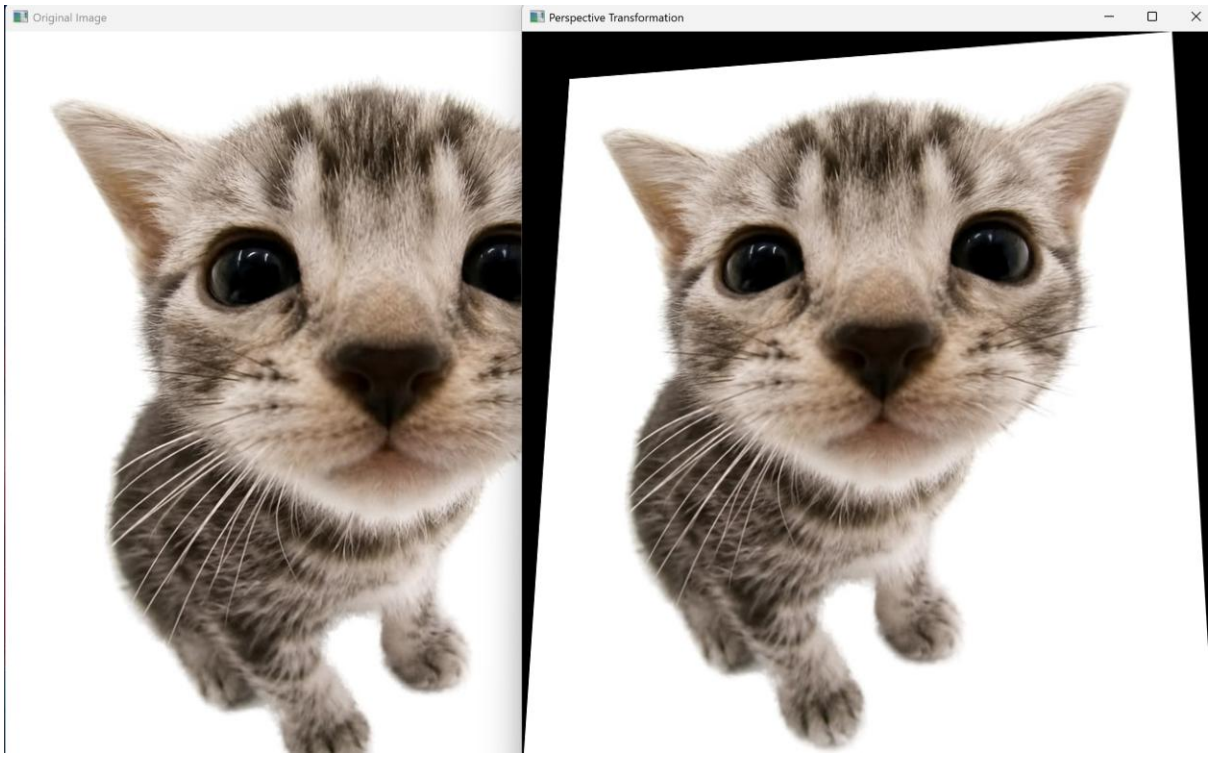
if img is not None:
    h, w = img.shape[:2]

    src = np.float32([[0, 0], [w-1, 0],
                      [0, h-1], [w-1, h-1]])
    dst = np.float32([[50, 50], [w-50, 0],
                      [0, h-50], [w-1, h-1]])

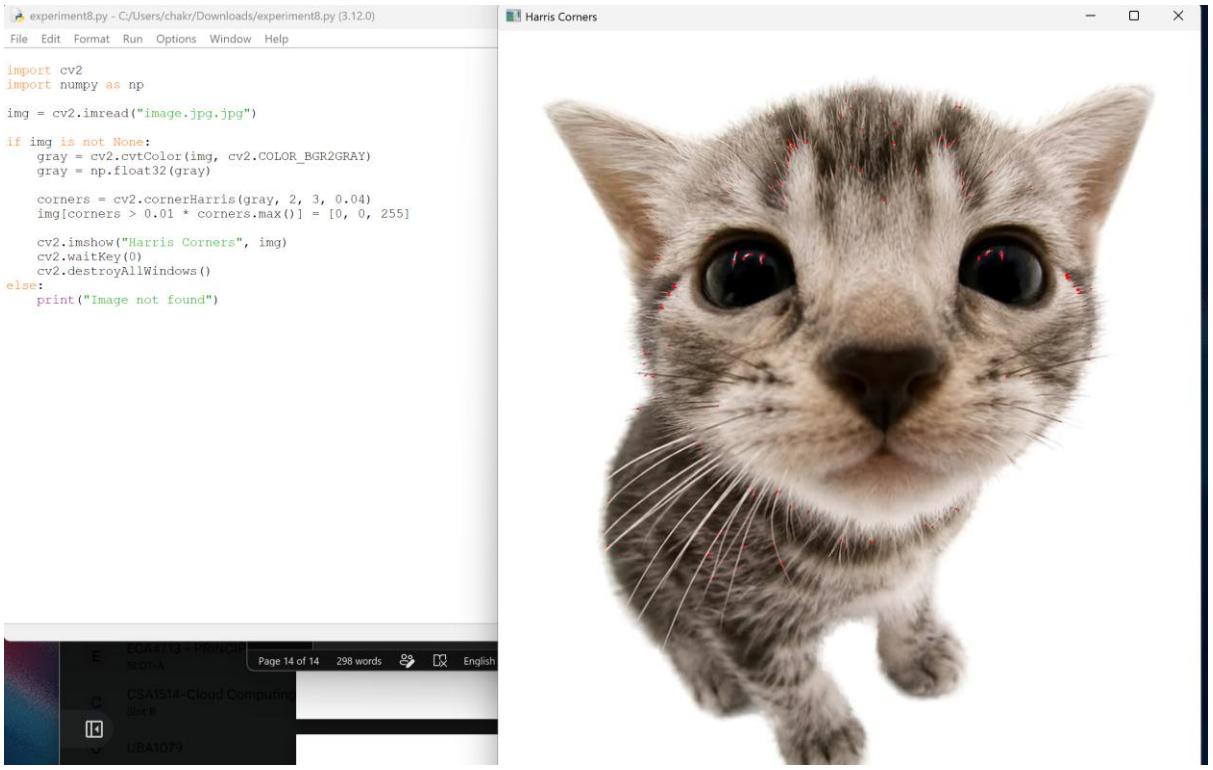
    M = cv2.getPerspectiveTransform(src, dst)
    result = cv2.warpPerspective(img, M, (w, h))

    cv2.imshow("Original Image", img)
    cv2.imshow("Perspective Transformation", result)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



15.Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function



16.Implement a Sobel algorithm using Open CV to filter the input image.

experiment8.py - C:/Users/chakr/Downloads/experiment8.py (3.12.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

if img is not None:
    x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
    y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

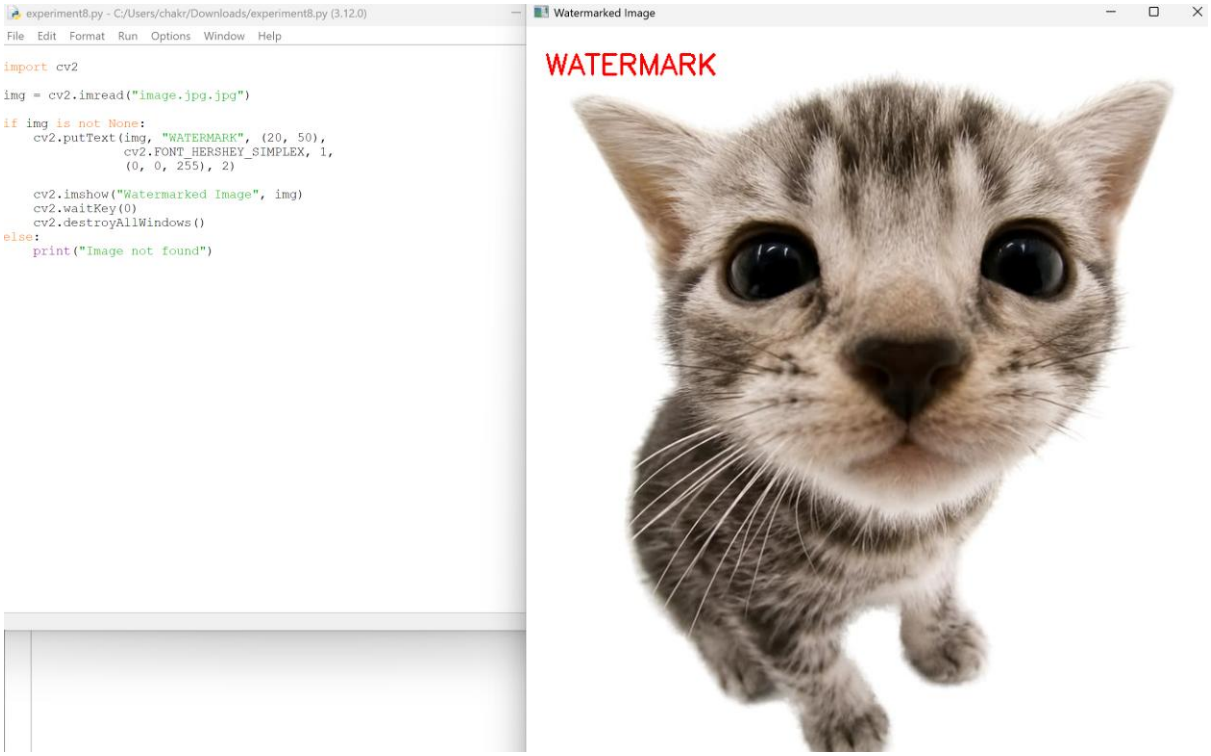
    edges = cv2.magnitude(x, y)
    edges = cv2.convertScaleAbs(edges)

    cv2.imshow("Original Image", img)
    cv2.imshow("Sobel Edge Detection", edges)

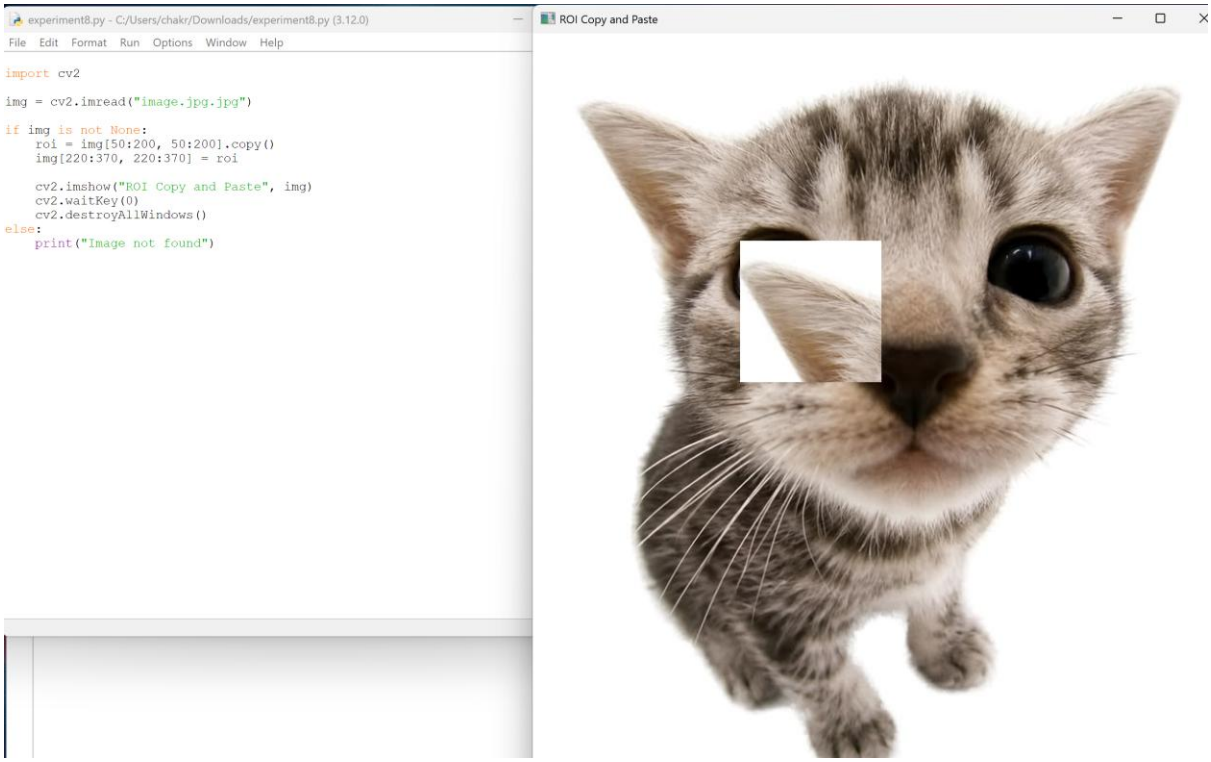
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



17.Design and implement a water marking technique to insert the watermark into the original effectively image using Open CV.



18.Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.



19.Implement the Erosion Morphological operations technique using Open CV in python.

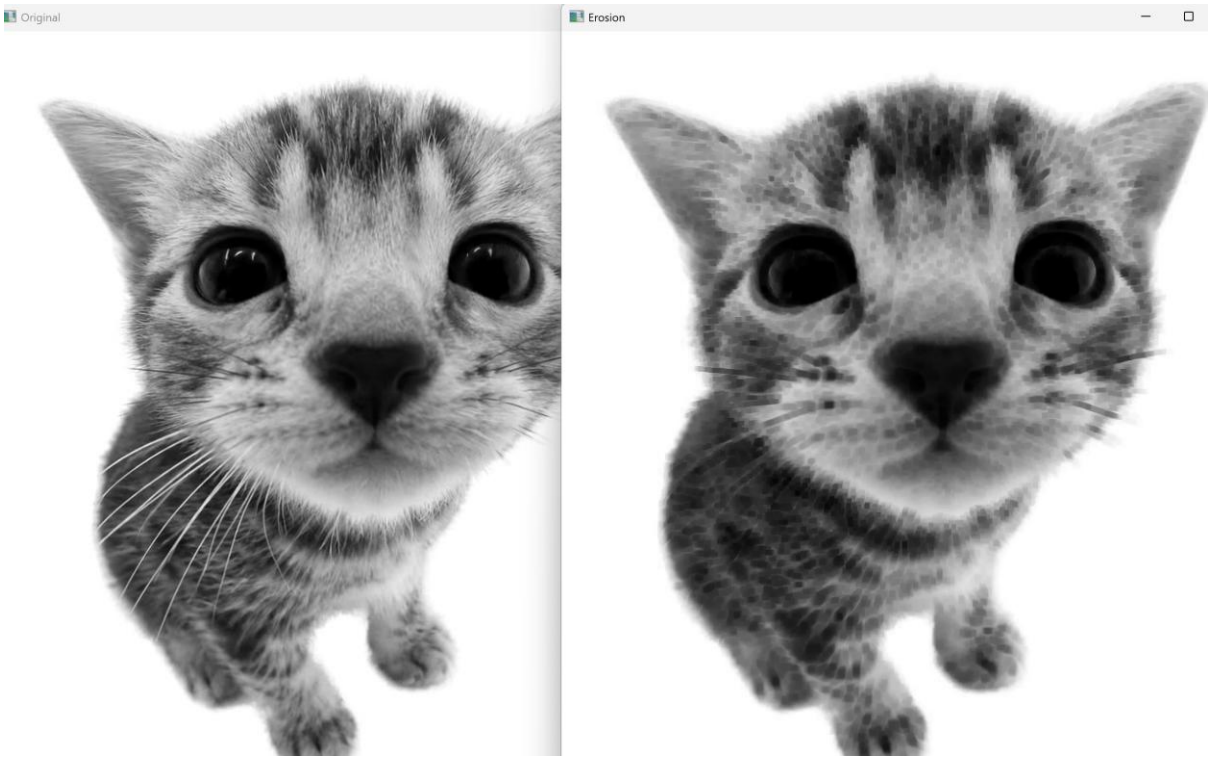
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.erode(img, kernel, iterations=1)

    cv2.imshow("Original", img)
    cv2.imshow("Erosion", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



20.Implement the Dilation technique as a Morphological operation to dilate the foreground regions based on Open CV.

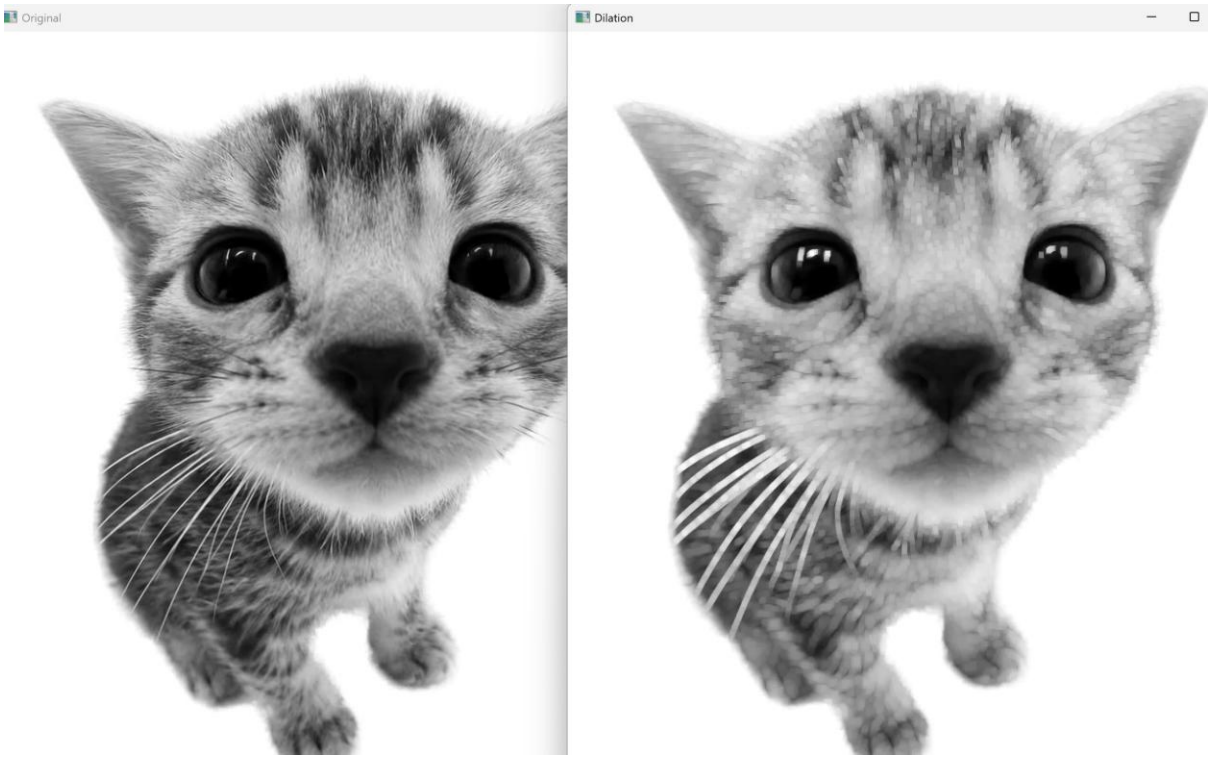
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.dilate(img, kernel, iterations=1)

    cv2.imshow("Original", img)
    cv2.imshow("Dilation", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



21.Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.

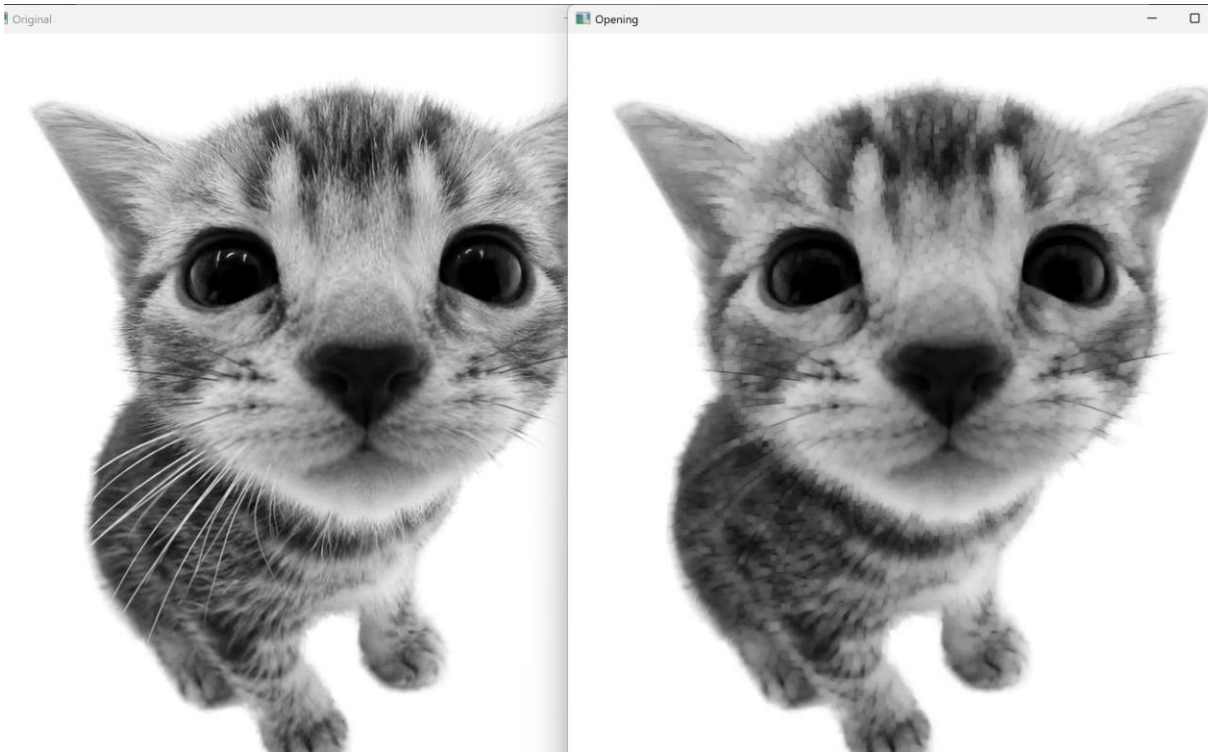
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)

    cv2.imshow("Original", img)
    cv2.imshow("Opening", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



22.Implement the Closing technique as a Morphological operation to dilate the foreground regions based on Open CV.

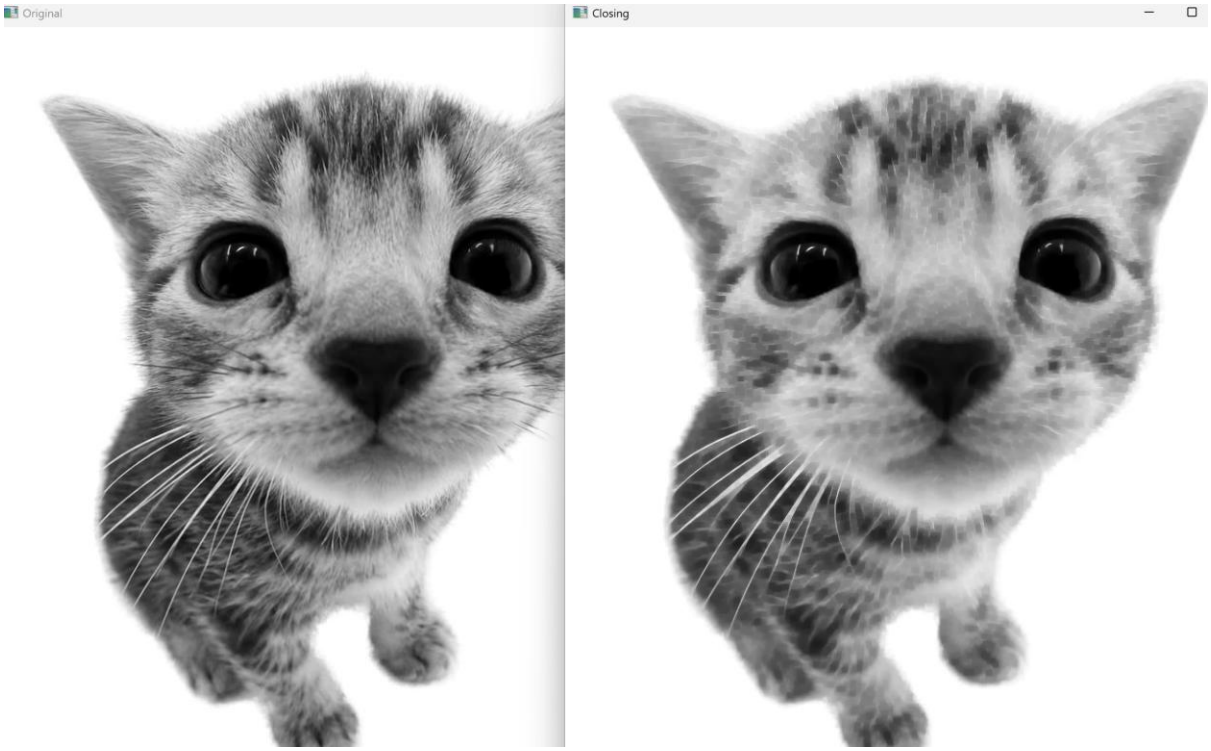
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpeg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

    cv2.imshow("Original", img)
    cv2.imshow("Closing", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



23.Implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)

    cv2.imshow("Original", img)
    cv2.imshow("Top Hat", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```



24.Implement the Black hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

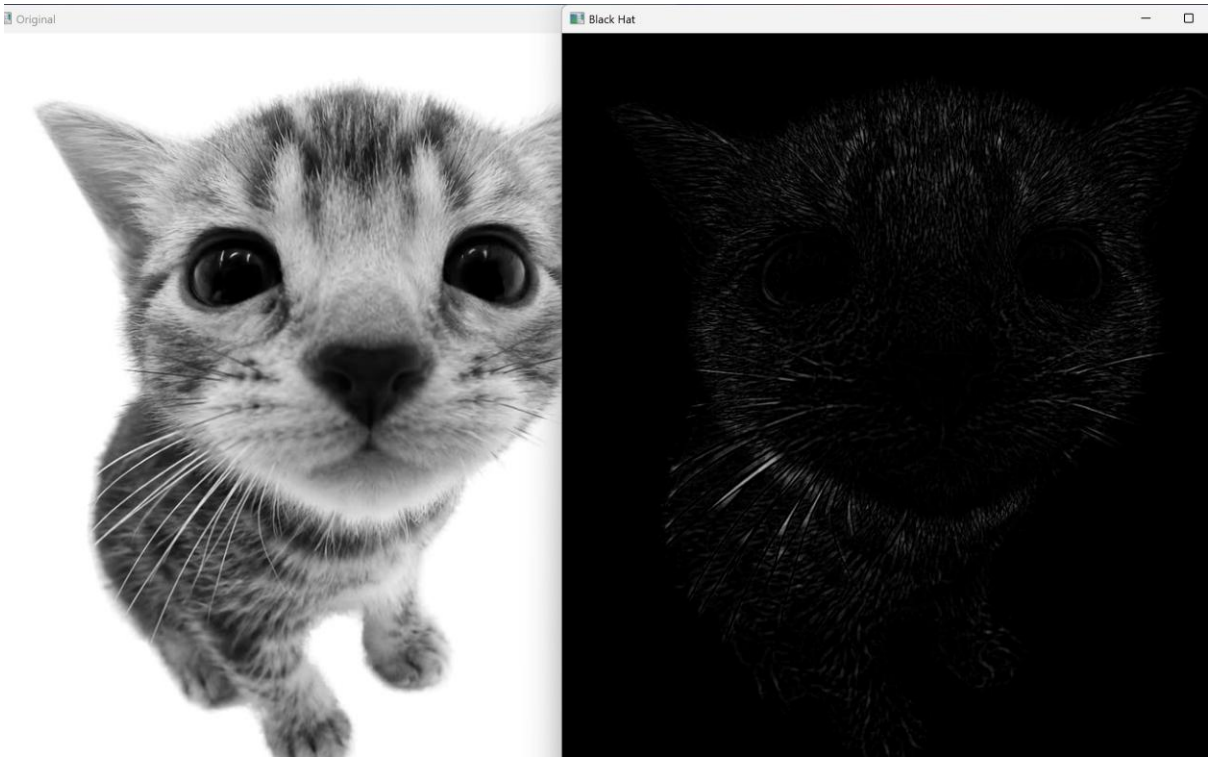
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread("image.jpg.jpeg", 0)

if img is not None:
    kernel = np.ones((5, 5), np.uint8)
    result = cv2.morphologyEx(img, cv2.MORPH_BLACKHAT, kernel)

    cv2.imshow("Original", img)
    cv2.imshow("Black Hat", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```


Original
Black Hat